The Biomechanics of the Push-up: Analysis of Joint Angles and Muscle Activation

Authors: Bret Contreras, MA, CSCS, et al.

Abstract & Overview

Source: American College of Sports Medicine (ACSM) Exercise Guidelines & Contreras Biomechanical Studies (2012)
Citation: Contreras, B., et al. (2012). The Biomechanics of the Push-up: Analysis of Joint Angles and Muscle Activation. Strength & Conditioning Journal, 34(5), 41-46.

1. Biomechanics of the Push-Up

The standard push-up is a closed-kinetic-chain horizontal pushing exercise activating the pectoralis major, anterior deltoids, triceps brachii, and rectus abdominis.
Key joint angles:
- Plank / Up Phase: Elbow angle is 160° to 180° (arms extended without joint jamming). The body line from shoulders to hips to ankles forms a rigid straight line (body angle > 160°).
- Bottom / Chest Phase: Elbow angle decreases to ≤ 90° as the chest approaches 2-3 inches from the ground.

2. Common Form Errors

• Error A: Hip Sagging (Lumbar Hyperextension)
  - Biomechanical Cause: The pelvis dips toward the floor (hip deviation > +0.08 normalized units), breaking the straight spinal alignment.
  - Root Causes: Weak transverse abdominis and gluteal muscles failing to resist lumbar extension under gravitational load.
  - Risk: Increases compressive shear load on lumbar vertebrae L4-S1.
  - Correction: Squeeze glutes and pull the belly button inward toward the spine; maintain a posterior pelvic tilt throughout the set.
• Error B: Piked Hips (Hips Too High)
  - Biomechanical Cause: The hips rise significantly above the shoulder-ankle line (hip deviation < -0.08 normalized units).
  - Root Causes: Compensatory mechanism to reduce core load by shortening the lever arm.
  - Correction: Lower the hips into a true neutral plank before initiating the descent.
• Error C: Elbow Flare
  - Biomechanical Cause: Flaring elbows out at 90° relative to the torso.
  - Risk: Increases subacromial impingement risk in the rotator cuff.
  - Correction: Keep elbows tucked at a 45° to 60° angle relative to the ribcage ("arrowhead" shape, not a "T" shape).

3. Recommended Camera Setup for Push-Ups

- Angle: Side (sagittal) view, placed at floor level or low tripod height (0.5 to 1 meter).
- Distance: 2.0 to 3.0 meters away.
- Framing: Full body length from head to heels must fit horizontally in the camera frame.